

Overview

Nginx as reverse proxy, TLS terminator, rate limiter, static file server, proxy cache, and Kubernetes Ingress. Event-driven architecture handles 10k+ connections with minimal memory (epoll, worker_processes auto).

nginx.conf Architecture

```
worker_processes auto; worker_connections 1024; (epoll eve
```

```
upstream shopverse_backend { least_conn; server app1:8080;
```

```
server block (port 80) → redirect to HTTPS
```

```
server block (port 443 ssl http2)
```

```
ssl_protocols TLSv1.2 TLSv1.3; Strict-Transport-Security
```

```
location /api/ → proxy_pass shopverse_backend; X-Real-IP,
```

```
location /images/ → alias /var/www/images/ (static files,
```

```
location = /health → return 200 OK (Nginx-level health che
```

```
location ~* \.(js|css|png)$ → far-future Expires header
```

Rate Limiting Zones

Zone	Key	Rate	Applied To
api	\$binary_remote_addr	60r/m	location /api/v1/orders (burst=20 nodelay)
login	\$binary_remote_addr	5r/m	location /api/v1/auth/login (burst=3 nodelay)

Proxy Cache Config

```
proxy_cache_path /var/cache/nginx levels=1:2 keys_zone=product_cache:10m max_size=1g
inactive=5m;
proxy_cache_valid 200 5m; proxy_cache_bypass $http_pragma;
add_header X-Cache-Status $upstream_cache_status;
```

Nginx vs Alternatives

Feature	Nginx	HAProxy	Traefik	Envoy
Model	Event-driven (epoll)	Event-driven	Event-driven	Event-driven
Config	nginx.conf (declarative)	haproxy.cfg	YAML / K8s CRD	YAML + xDS API
Static file serving	Excellent	No	No	No
K8s Ingress	Nginx Ingress Controller	HAProxy Ingress	Native	Istio / Ambassador

Feature	Nginx	HAProxy	Traefik	Envoy
Dynamic config reload	nginx -s reload	Runtime (no reload)	Hot reload	xDS (live)
Learning curve	Low	Medium	Low	High
ShopVerse use	Primary	—	—	—

Kubernetes Ingress Annotations

```
nginx.ingress.kubernetes.io/rate-limit: "100"
```

```
nginx.ingress.kubernetes.io/proxy-body-size: "10m"
```

```
nginx.ingress.kubernetes.io/enable-cors: "true"
```

```
nginx.ingress.kubernetes.io/rewrite-target: /$2
```